

Automated Warehouse Control System

Micro850 PLC (CCW) + Factory I/O | Conveyors + Crane (Lift + Forks)

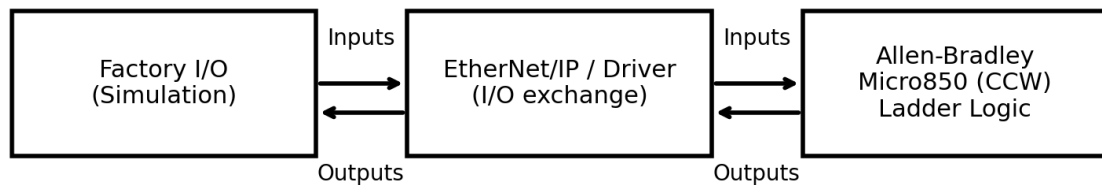
Executive Summary

This project implements an automated infeed, storage, and outfeed workflow in Factory I/O, controlled by Allen-Bradley Micro850 ladder logic. The PLC sequences conveyors, emitter, lift, and fork actuators using a step-based state machine with safety interlocks, sensor inversion handling, timers/edge-detect logic, and deterministic load/unload cycles.

Tech Stack

- Allen-Bradley Micro850 (simulation)
- Connected Components Workbench (CCW)
- Factory I/O (Automated Warehouse scene)
- EtherNet/IP driver and discrete I/O mapping

System Architecture



I/O Mapping (Factory I/O → PLC Tags)

Signal	Type	Factory I/O	PLC (BOOL_IN/OUT)
At Entry	Input	At Entry sensor	BOOL_IN_0
At Load	Input	At Load sensor	BOOL_IN_1
At Left	Input	At Left sensor	BOOL_IN_2
At Middle	Input	At Middle sensor	BOOL_IN_3
At Right	Input	At Right sensor	BOOL_IN_4
At Unload	Input	At Unload sensor	BOOL_IN_5
At Exit	Input	At Exit sensor	BOOL_IN_6
Moving X	Input	Crane moving X	BOOL_IN_7
Moving Z	Input	Crane moving Z	BOOL_IN_8
Start PB	Input	Start	BOOL_IN_9
Reset PB	Input	Reset	BOOL_IN_10
Stop PB	Input	Stop	BOOL_IN_11
E-Stop	Input	Emergency stop	BOOL_IN_12
Auto Mode	Input	Auto selector	BOOL_IN_13
Entry Conveyor	Output	Entry Conveyor	BOOL_OUT_0
Load Conveyor	Output	Load Conveyor	BOOL_OUT_1
Forks Left	Output	Forks Left actuator	BOOL_OUT_2
Forks Right	Output	Forks Right actuator	BOOL_OUT_3
Lift	Output	Lift actuator	BOOL_OUT_4
Unload Conveyor	Output	Unload Conveyor	BOOL_OUT_5
Exit Conveyor	Output	Exit Conveyor	BOOL_OUT_6
Start Light	Output	Start light	BOOL_OUT_7
Stop Light	Output	Stop light	BOOL_OUT_9

Emitter

Output

Emitter

BOOL_OUT_10

Note: Some Factory I/O sensors are 'normally ON' (clear=1, blocked=0). In the ladder, create *_Box tags using NOT to normalize 'Box Present = 1' semantics.

Control Strategy

- Step-based state machine (integer Step tag) for deterministic sequencing
- Separate Load_Sequence_Active and Unload_Sequence_Active latches
- Interlocks: Auto_Enable, RunCmd, E-Stop OK, mutual exclusion between load/unload
- Edge detection (emitter pulse, motion completion)
- MAX_SHELF and Stored_Shelf_Value to scale storage positions

Load Sequence

- 1) Conveyors move box to AtLoad
- 2) Forks extend to pick
- 3) Lift up
- 4) Forks retract
- 5) Move X to target shelf (Stored_Shelf_Value)
- 6) Lift to shelf height
- 7) Forks extend to place
- 8) Forks retract
- 9) Return crane to Home (X=HomePos, Z=Down)
- 10) Clear load sequence, increment Stored_Shelf_Value

Unload Sequence

- 20) Move X to selected shelf
- 21) Lift to shelf height
- 22) Forks extend to pick
- 23) Forks retract
- 24) Return to unload position
- 25) Lift down
- 26) Forks extend to place on outfeed
- 27) Forks retract
- 28) Convey to Exit
- 29) Clear unload sequence

Appendix: Key Screenshots

Factory I/O I/O mapping view:

Host: 10.0.0.70			
At Entry	BOOL_IN_0	BOOL_OUT_0	Entry Conveyor
At Load	BOOL_IN_1	BOOL_OUT_1	Load Conveyor
At Left	BOOL_IN_2	BOOL_OUT_2	Forks Left
At Middle	BOOL_IN_3	BOOL_OUT_3	Forks Right
At Right	BOOL_IN_4	BOOL_OUT_4	Lift
At Unload	BOOL_IN_5	BOOL_OUT_5	Unload Conveyor
At Exit	BOOL_IN_6	BOOL_OUT_6	Exit Conveyor
Moving X	BOOL_IN_7	BOOL_OUT_7	Start light
Moving Z	BOOL_IN_8	BOOL_OUT_8	Start light
Start	BOOL_IN_9	BOOL_OUT_9	Stop light
Reset	BOOL_IN_10	BOOL_OUT_10	Emitter
Stop	BOOL_IN_11	BOOL_OUT_11	
Emergency stop	BOOL_IN_12	BOOL_OUT_12	
Auto	BOOL_IN_13	BOOL_OUT_13	
FACTORY I/O (Running)	BOOL_IN_14	BOOL_OUT_14	
Manual	BOOL_IN_15	BOOL_OUT_15	
	BOOL_IN_16	BOOL_OUT_16	
	BOOL_IN_17	BOOL_OUT_17	
	BOOL_IN_18	BOOL_OUT_18	
	BOOL_IN_19	BOOL_OUT_19	
	BOOL_IN_20	BOOL_OUT_20	
	BOOL_IN_21	BOOL_OUT_21	
	BOOL_IN_22	BOOL_OUT_22	
	BOOL_IN_23	BOOL_OUT_23	
	BOOL_IN_24	BOOL_OUT_24	
	BOOL_IN_25	BOOL_OUT_25	
	BOOL_IN_26	BOOL_OUT_26	
	BOOL_IN_27	BOOL_OUT_27	
	BOOL_IN_28	BOOL_OUT_28	
	BOOL_IN_29	BOOL_OUT_29	
	BOOL_IN_30	BOOL_OUT_30	
	BOOL_IN_31	BOOL_OUT_31	

